GENERALIZATION OF TOPMODEL FOR A POWER LAW TRANSMISSIVITY PROFILE

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ABSTRACT

A generalization of the TOPMODEL equations for a power law vertical profile of hydraulic conductivity is introduced. The exponential profile of TOPMODEL is obtained as a limit case of the new general form.

Hydrol. Process., Vol. 11, 1353–1355 (1997)

(No. of Figures: 1 No. of Tables: 0 No. of Refs: 6)

KEY WORDS TOPMODEL; topographic index; power law transmissivity profile; recession analysis

INTRODUCTION

Recently Ambroise *et al.* (1996, see also Beven, 1997) presented a generalization of the TOPMODEL equations for the cases of parabolic and linear profiles of downslope transmissivity, as an extension of the original exponential assumption. Their justification for this extension was based on an analysis of recession curves for the Ringelbach catchment that did not show the first-order hyperbolic form in time to be expected from an exponential storage–discharge function. In this contribution we develop the TOPMODEL equations for a general power law profile. The linear and parabolic cases of Ambroise *et al.* represent particular cases $n = 1$ and $n = 2$ in what follows.

MODEL EQUATIONS

Figure 1 presents, in a parallel format, TOPMODEL equations derived from two forms of downslope transmissivity profile: the exponential profile used in the original TOPMODEL and the power law profile [see also Figure 2 of Ambroise *et al.* (1996)]. Similar equations can be obtained when the water table depth is used instead of saturation deficit as an independent variable. A full mathematical derivation is given by Iorgulescu (1997). The equations are obtained by applying the same assumptions to the two forms of transmissivity profiles given by Equation (1) of Figure 1. These assumptions are:

- A1 The dynamics of the saturated zone can be approximated by successive steady-state representations.
- A2 The hydraulic gradient of the saturated zone is approximated by the local surface topographic slope ($\tan \beta$).
- A3 The shape parameters of the hydraulic conductivity profile [m or M and n in Equation (1)] are constant in space.
- A4 The saturated zone receives a spatially uniform recharge rate (q_v).

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Contract grant sponsor: OFES-EV.
Contract grant number: 95-00449.

CCC 0885–6087/97/091353–03$17·50

Received 13 February 1997
Revised 3 March 1997
Accepted 25 March 1997

Type of transmissivity profile

exponential law | **power law**

$$T(D_i) = To_i \exp(-D_i/m) \qquad\qquad T(D_i) = To_i\,(1-D_i/M)^n \qquad (1)$$

where To_i [$L^2\ T^{-1}$] is the lateral transmissivity when the soil is just saturated, D_i [L] is the local saturation storage deficit, m [L] is a parameter which may be interpreted as an effective storage capacity, M [L] is the maximum gravity storage of the permeable layer and n [-] is an exponent.

Under the assumption A2 the downslope discharge at any point in the catchment is given by :

$$q_i = To_i \tan\beta_i \exp(-D_i/m) \qquad\qquad q_i = To_i \tan\beta_i (1-D_i/M)^n \qquad (2)$$

With assumptions A1 and A4 local downslope flow is $q_i = q_v\, a_i$, where a_i is the upslope area draining through this point per contour length. The mean storage deficit (D) is defined by averaging local saturation deficits over the total catchment area (A) and using A3. The relationship between local and mean storage deficit is :

$$D_i/m - D/m = G_{\infty i} - \gamma_\infty \qquad\qquad (1-D_i/M)/(1-D/M) = G_{ni}/\gamma_n \qquad (3)$$

where $G_{\infty i}$ and G_{ni} are the soil-topographic indexes defined by the exponential and power profile respectively and γ_∞ and γ_n are their spatial averages :

$$G_{\infty i} = \ln\left(a_i/(\tan\beta_i To_i)\right) \qquad\qquad G_{ni} = \left(a_i/(\tan\beta_i To_i)\right)^{1/n} \qquad (4)$$

$$\gamma_\infty = A^{-1}\int_A \ln\left(a_i/(\tan\beta_i To_i)\right) dA \qquad\qquad \gamma_n = A^{-1}\int_A \left(a_i/(\tan\beta_i To_i)\right)^{1/n} dA \qquad (5)$$

Total drainage from the saturated zones $Q_b(D)$ [$L^3\ T^{-1}$] is calculated by summing subsurface flows along channel reaches :

$$Q_b(D) = A\exp(-\gamma_\infty)\exp(-D/m) \qquad\qquad Q_b(D) = A(\gamma_n)^{-n}(1-D/M)^n \qquad (6)$$

Under the assumption of an uniform downslope transmissivity at saturation (To), the topographic index (l) expressions and their spatial averages (λ) are defined by :

$$l_{\infty i} = \ln\left(a_i/\tan\beta_i\right) \qquad\qquad l_{ni} = \left(a_i/\tan\beta_i\right)^{1/n} \qquad (7)$$

$$\lambda_\infty = A^{-1}\int_A \ln\left(a_i/\tan\beta_i\right) dA \qquad\qquad \lambda_n = A^{-1}\int_A \left(a_i/\tan\beta_i\right)^{1/n} dA \qquad (8)$$

Figure 1. TOPMODEL relationships for two types of downslope transmissivity profile

Furthermore, making the notation $m^* = M/n$, the exponential profile of the original TOPMODEL [Equation (1)] can be obtained as a limit case of the power law, when n goes to infinity (e.g. Piskunov, 1981, pp. 49–51):

$$\lim_{n\to\infty}(\mathrm{To}_i(1 + (-D_i/m^*)/n)^n) = To_i \exp(-D_i/m^*) \qquad (9)$$

In order that the m^* parameter be finite, the storage of the saturated zone M should also go to infinity. This is consistent with the hypotheses of the TOPMODEL exponential law. Thus, taking the relations for the power law profile to the limit ($n \to \infty$) leads to results identical with those obtained directly from the exponential profile. Relationships between the classes of soil topographic [Equation (4)] or topographic indexes as a function of n are immediate and a topographic analysis needs to be performed only once.

RECESSION ANALYSIS

Brutsaert and Nieber (1977) present an analysis technique for recession curves having the power form of Equation (10) where a and B are parameters:

$$\mathrm{d}Q_b/\mathrm{d}t = -aQ_b^B \tag{10}$$

It can be shown that the recessions of the generalized TOPMODEL subsurface store [Equation (6)] follow Equation (10), where the relationship between parameter n in Equation (1) and parameter B in Equation (10) is given by:

$$B = (2n - 1)/n \tag{11}$$

For values of n between one (linear transmissivity profile) and infinity (exponential transmissivity profile) B takes values between one (linear transmissivity profile) and two (exponential transmissivity profile). This allows the n parameter to be estimated, and also the adequacy of Equation (6) for the modelling of subsurface flows to be tested.

This possibility is briefly exemplified here by using this method for several sub-basins of the Haute–Mentue (Switzerland) research basin. The exponents B derived from individual recessions were found to be variable in time and space, but generally lie between 2·5 and 3·5 (Iorgulescu, 1997). This is outside the theoretical range for these steady-state relationships and if used to calculate an effective value of n would result in negative values. The exponential profile ($B = 2$) should give the best simulation results for this catchment, as it has the closest value to the observed ones. However, the calculated B suggests that assumption A1 may be violated. As Brutsaert and Nieber (1977) show, a value of $B = 3$ corresponds to the short time solution of the Boussinesq equation. This could indicate that a model that takes more account of transient processes may be required to explain adequately the observed behaviour of these catchments (see also Kirkby, 1997, for an analysis of the quasi-steady-state assumption).

SUMMARY

The TOPMODEL equations are generalized for a power law profile of downslope transmissivity. With respect to previous contributions this increases the flexibility in modelling different forms of recession and variation of hydraulic conductivity with depth. A class of topographic indices, appropriate for the new mathematical description, is also introduced. The TOPMODEL exponential law is found as a limit form of the present equations. The recession analysis method of Brutsaert and Nieber (1977) is suggested for a first estimation of the profile form and as a test for the assumptions adopted for the behaviour of the subsurface store.

ACKNOWLEDGEMENTS

This research was supported by OFES Grant # 95.00449 in the framework of the European Union VAHMPIRE Project. Prof. K. J. Beven is gratefully acknowledged for his highly stimulating discussions and comments.

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